

Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - II

Course Code : CSE 0533 1201 (CSE) & PHY 0533 1201 (EEE)

Term: Spring – 2024	Date: 08.01.2024	Time: 03.30 PM	Room: 327	Activity: Lecture 3
----------------------------	-------------------------	-----------------------	------------------	----------------------------

Topics to be discussed: Kinetic theory of gases & Its applications

Mean free path, Equipartition of energy & Brownian motion

Target CLO & PLO : CLO2-3 & PLO 2 & 5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define elastic collision, free path & mean free path
- ❖ explain the Brownian motion
- ❖ explain the Equipartition of energy & Define Degree of freedom
- ❖ solve assigned recommended mathematical problems

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

- 1) What is the unit of the mean free path?
a) N b) m c) F d) Nm^{-1} e) mN^{-1}
- 2) The degree of freedom of a non-linear triatomic gas is?
a) 3 b) 5 c) 6 d) 7 e) 9
- 3) The mean free path is proportional to k^{th} power of the diameter of the molecules. Here value of k is –
a) -2 b) -1 c) 0 d) 1 e) 2

Sample Questions (SAQ):

1. Define free path & mean free path.
2. Write short notes about Brownian motion.
3. Write short notes about “Equipartition of energy”.
4. Write short notes about “Maxwell’s distribution of molecular speeds”
5. What do you mean by Degree of Freedom?
6. Derive the equation for the Mean Free Path of gas molecules.
Or, Show that, the Mean free path is inversely proportional to its density.
Or, Prove, $\lambda = \frac{1}{\pi a^2 n}$; Here the symbols have their normal meaning.
Or, What are the factors on which mean free depends?
7. The diameter of the molecule of a gas is $3 \times 10^{-10}\text{m}$ and number of molecules per cm^3 is 6×10^{20} . Calculate the mean free path of the molecules.
8. The mean free path of the molecules of a gas is $6 \times 10^{-8}\text{m}$ and the diameter of a molecule is $2.5 \times 10^{-10}\text{m}$. Find the number of molecules per m^3 .

Reference Books:

1. Giasuddin ahmed – Physics for Engineers (Part-1)
2. Halliday Resnick - Fundamental of Physics, 2nd edition

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 01-64
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 01-36